

# **Microscopic Traffic Simulation: Calibration and Dynamic applications**

Task 1: Model Calibration

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# Overview

1. Simulation replications
2. Measure of Performance
3. Input space exploration
4. Hyperparameter optimization
5. Early SPSA results

# SIMULATION REPLICATIONS

# On simulation stochasticity

Microscopic simulations with SUMO are inherently stochastic: the simulator uses RNGs with different (configurable) seeds to simulate random human behavior (rerouting probability, driving styles, route choice).

This reflects in the results of the simulation themselves: a simulation run with the same exact parameters will yield different results based on the RNG seed.

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How do we deal with this?

# Required simulation number

We **average** multiple simulation results. The number of simulations required depends on the tolerance and confidence level we want to achieve.

$$n = \left( \frac{t_{\alpha/2, n-1} \cdot \sigma}{W} \right)^2$$

where:

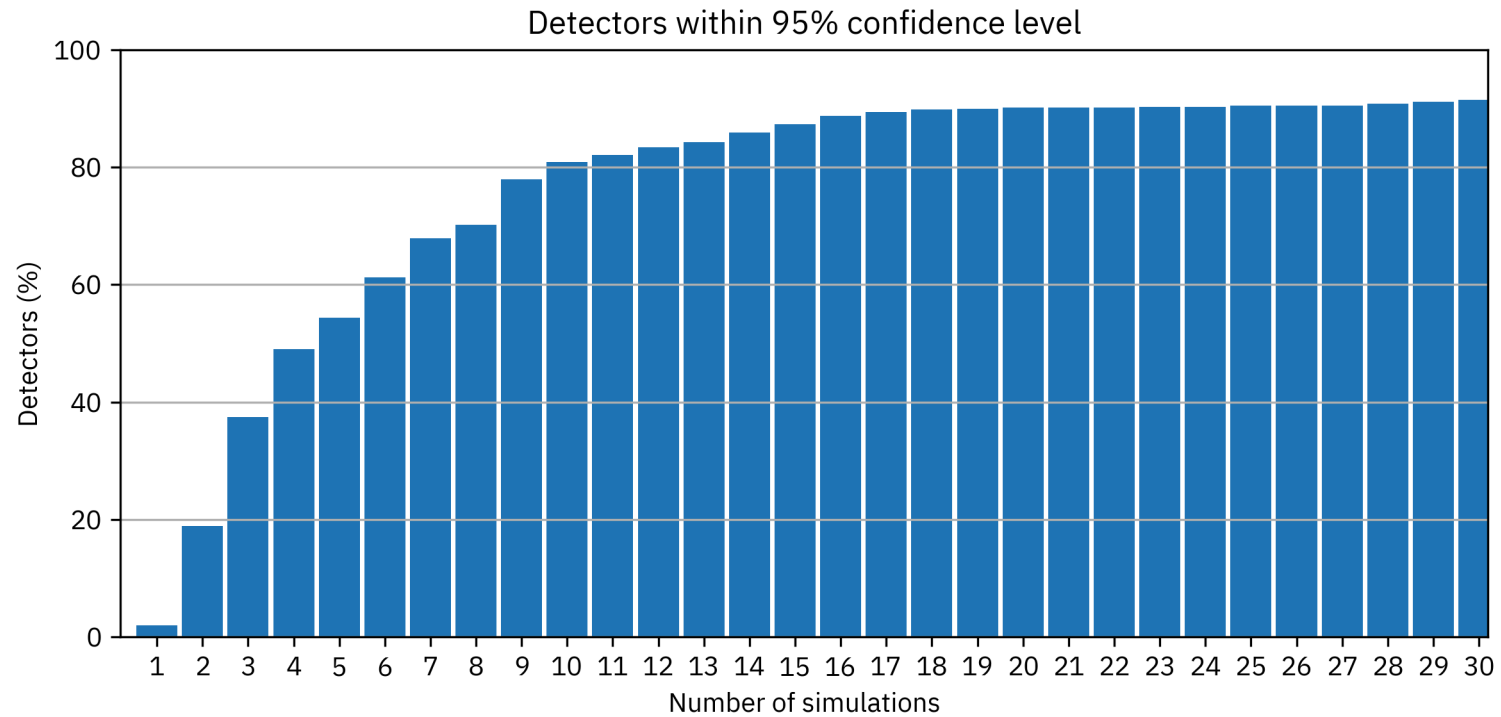
$n$  number of required simulations

$t$  t-students statistic (for  $\alpha/2$  significance,  $n - 1$  DOF)

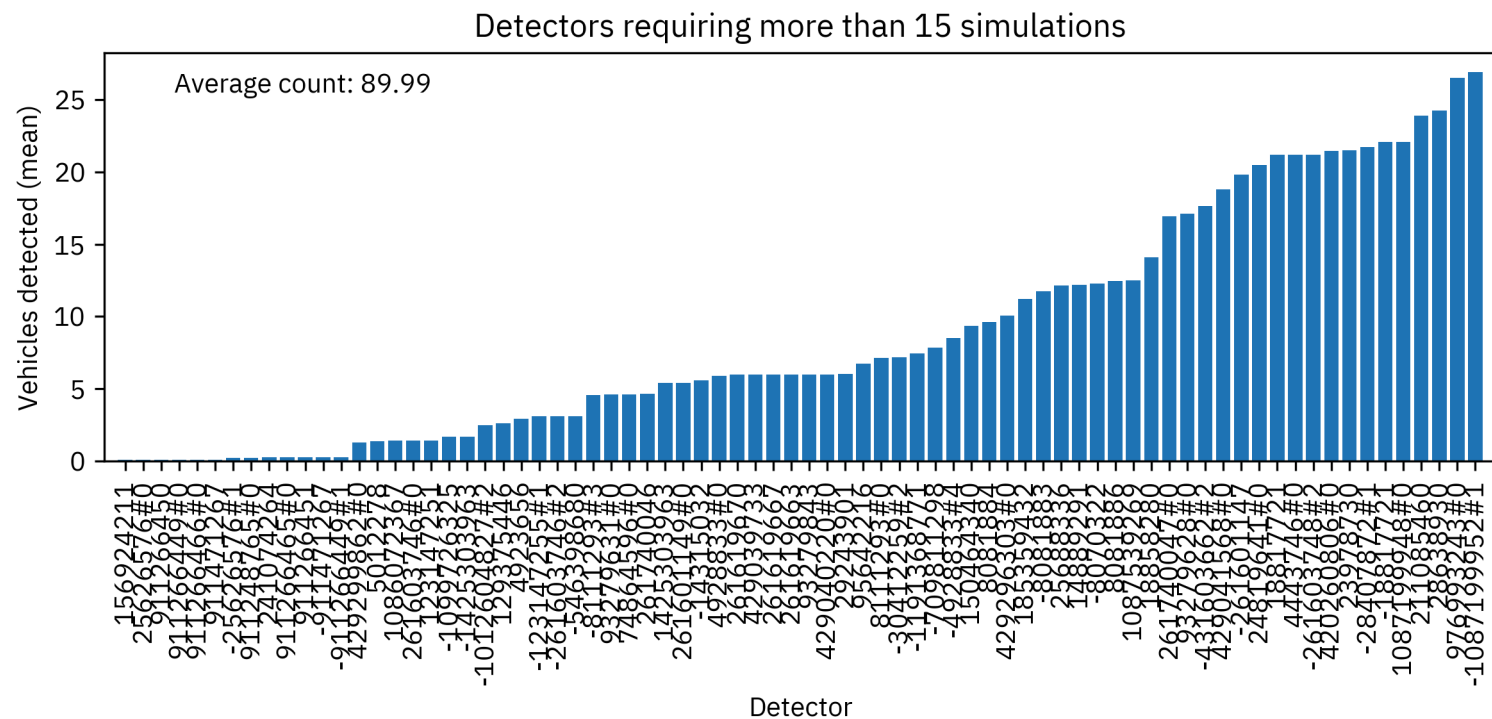
$\sigma$  standard deviation

$W$  width of tolerance interval

# Required simulation number



## Excluded detectors



## Simulation replications

MEASURE OF PERFORMANCE

# Limits of common objective functions

Some common objective functions, like MAPE and RMSE, show their limits when applied to a traffic calibration problem.

## RMSE

- Absolute metric: can put errors of different significance on the same scale (e.g. 100 vehicles on an Autobahn / Residential street)
- Dominated by outliers (as errors scale quadratically)

## MAPE

- Unstable on small flows: when the denominator is small, MAPE can explode
- Asymmetric

# Benefits of RMSN

- **Independent from scale:** allows to compare GOF across networks / traffic conditions
- **Normalization:** weights the RMSE against the global mean of observations, avoiding both instability on small flows and outlier dominance
- **Interpretability:** can be interpreted as an estimate of the coefficient of variation and expressed as a percentage

# GLS-based weighting schemes

Generalized least squares weight errors based on the variance of observations, using the inverse of the variance-covariance matrix  $V$ :

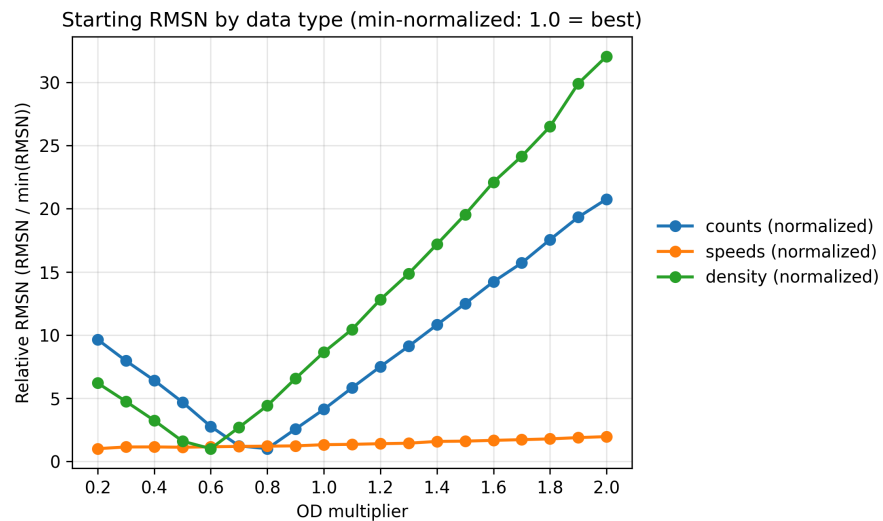
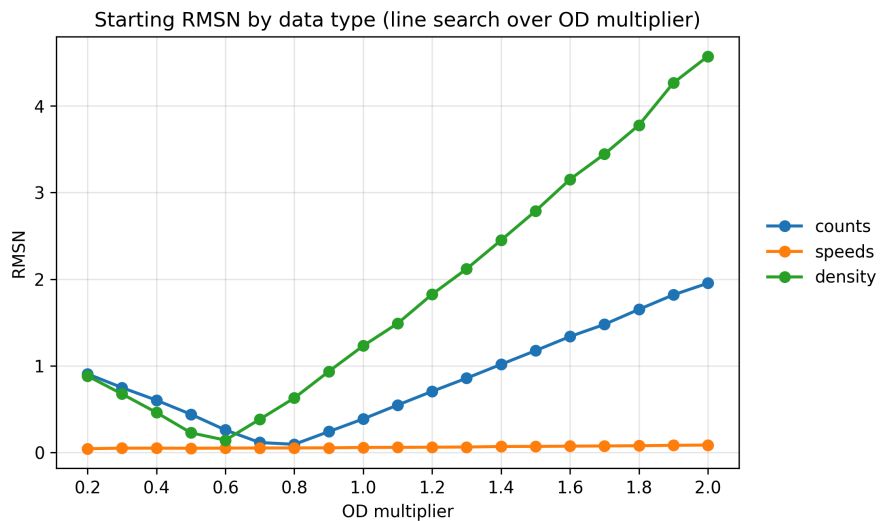
$$\text{GLS} = (Y_{\text{sim}} - Y_{\text{true}})^T V^{-1} (Y_{\text{sim}} - Y_{\text{true}})$$

The use of the matrix  $V$  allows to account for both heteroscedasticity and spatial correlation of observations.

**However**, in the absence of an a priori information on  $V$ , we would need to estimate it from data.

# INPUT SPACE EXPLORATION

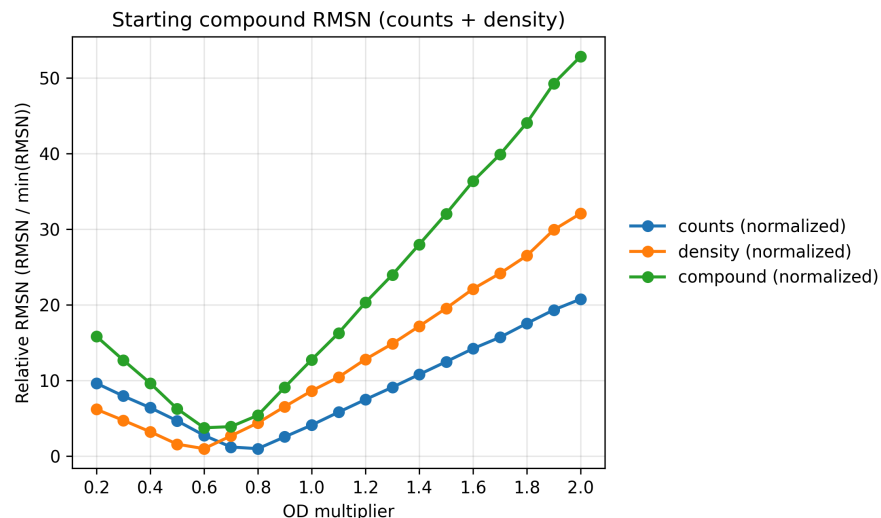
# Line search on starting RMSN



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As the speed RMSN seems to not be particularly affected by changes in the OD structure, it was given a very low weight in the final compound measure.

$$w_c, w_d \in (0.3, 0.6), w_s = 0.1$$



# HYPERPARAMETER OPTIMIZATION

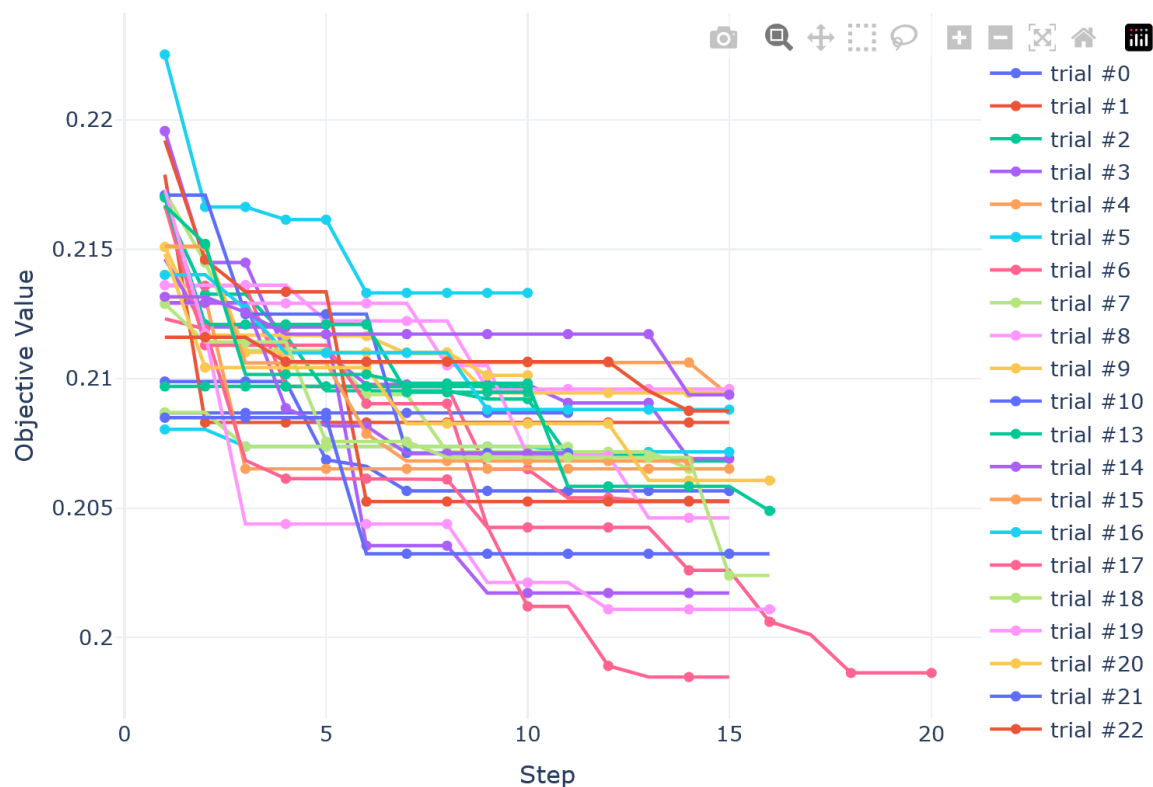
# Hyperparameter optimization with OPTUNA

Three SPSA parameters where optimized using the OPTUNA library:

- Initial step size  $a$
- Perturbation size  $c$
- Stability constant  $A$

OPTUNA allows to perform Bayesian optimization of a model hyperparameters, which leads to a much more efficient exploration of the parameter space compared to grid or random search.

# Results of hyperparameter optimization

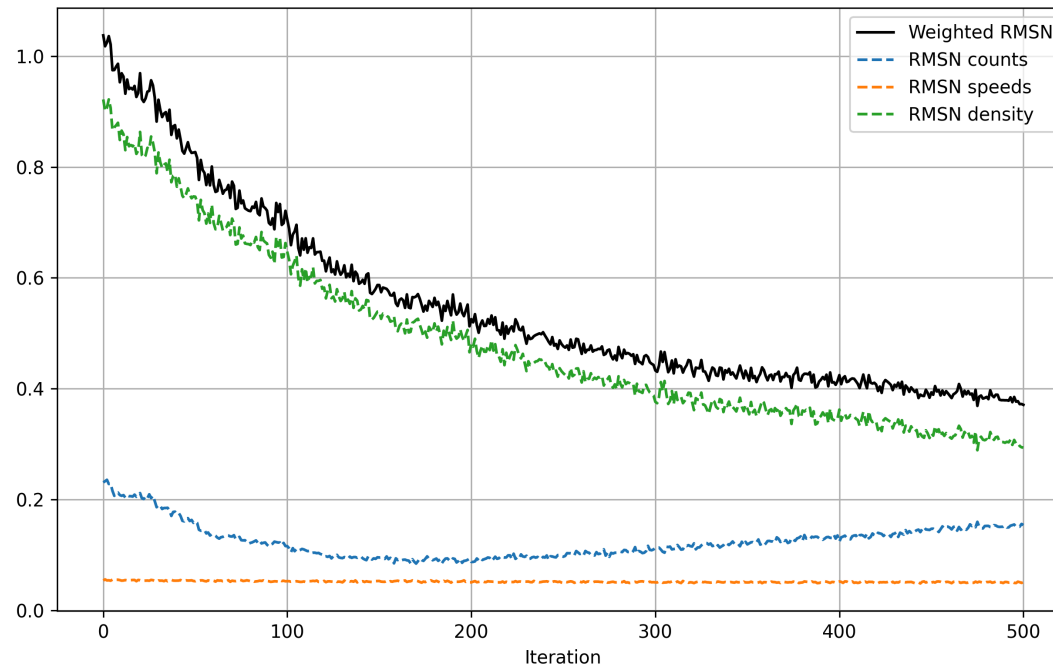


Final parameter selection:

- $a = 100$
- $c = 0.6$
- $A = 40$

# EARLY SPSA RESULTS

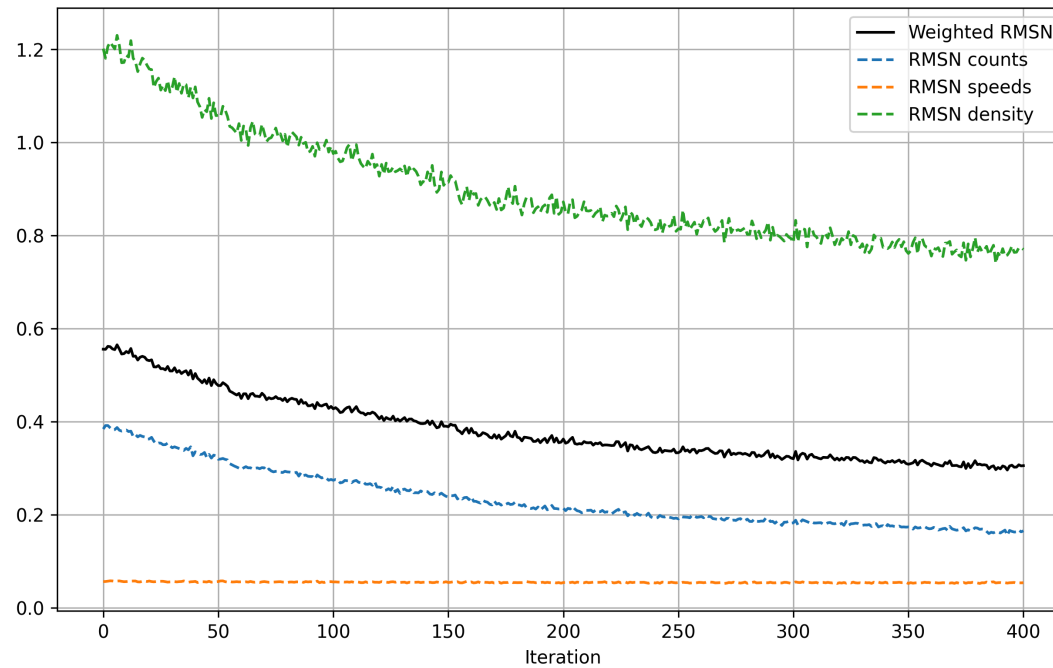
# RMSN components progression



Weights:

- $w_c = 0.3$
- $w_d = 0.6$
- $w_s = 0.0$

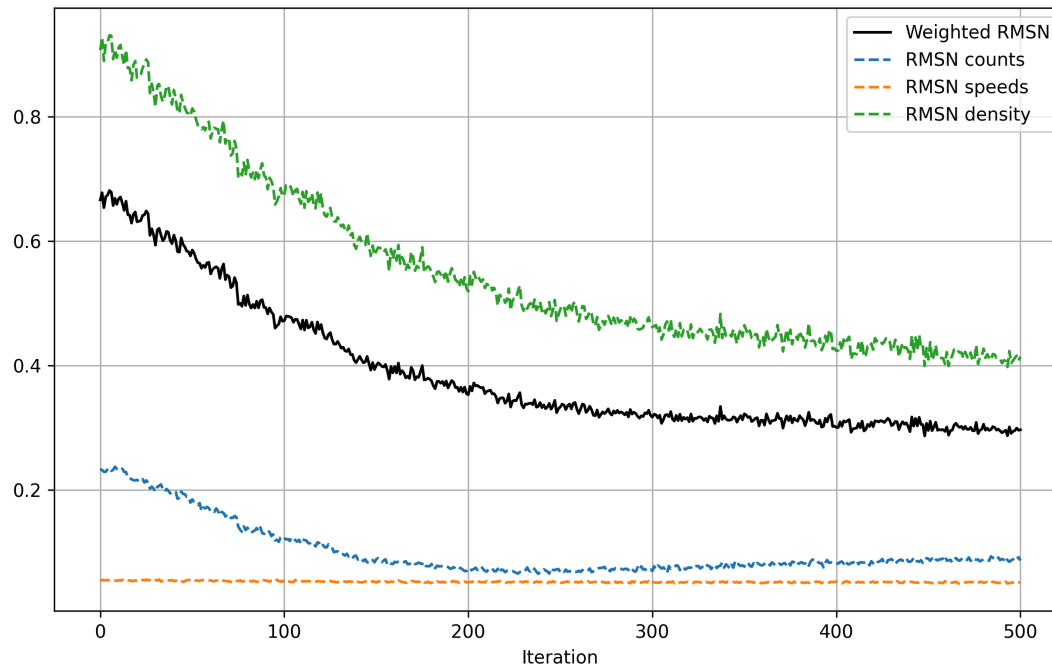
# RMSN components progression



Weights:

- $w_c = 0.5$
- $w_d = 0.2$
- $w_s = 0.1$

# RMSN components progression

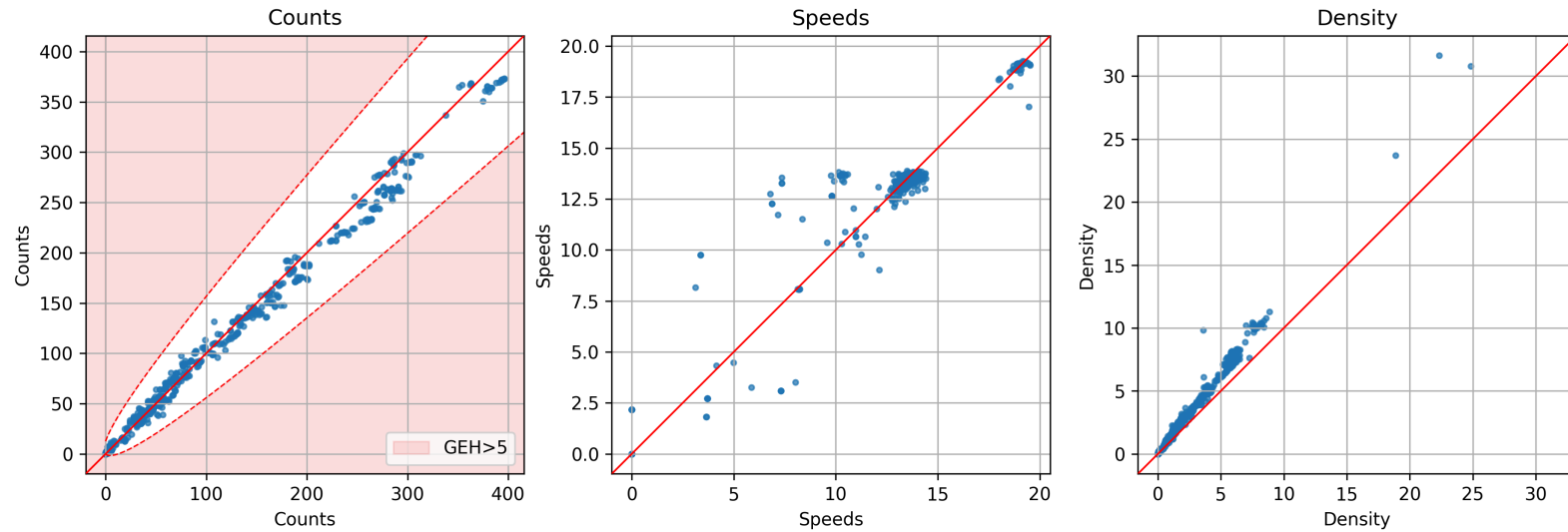


Weights:

- $w_c = 0.5$
- $w_d = 0.6$
- $w_s = 0.1$

# Predicted vs Actual flows

Predicted vs Actual (best RMSN at iteration 448)



Thank you!